

FORM TP 2012022



TEST CODE **01238020**

JANUARY 2012

C A R I B B E A N E X A M I N A T I O N S C O U N C I L

**SECONDARY EDUCATION CERTIFICATE
EXAMINATION**

PHYSICS

Paper 02 – General Proficiency

2 hours 30 minutes

READ THE FOLLOWING INSTRUCTIONS CAREFULLY.

1. This paper consists of SIX questions.
2. Section A consists of THREE questions. Candidates must attempt ALL questions in this section. Answers for this section must be written in this answer booklet.
3. Section B consists of THREE questions. Candidates must attempt ALL questions in this section. Answers for this section must be written in this answer booklet.
4. All working MUST be CLEARLY shown.
5. The use of silent, non-programmable calculators is permitted, but candidates should note that the use of an inappropriate number of figures in answers will be penalised.
6. Mathematical tables are provided.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO.

SECTION A

Attempt ALL questions.

You MUST write your answers in this answer booklet.

1. A Form 4 Physics student performed an activity to determine the mass of a lump of plasticine, M_p . A diagram of the setup of the apparatus used is seen in Figure 1 below. The Principle of Moments was used.

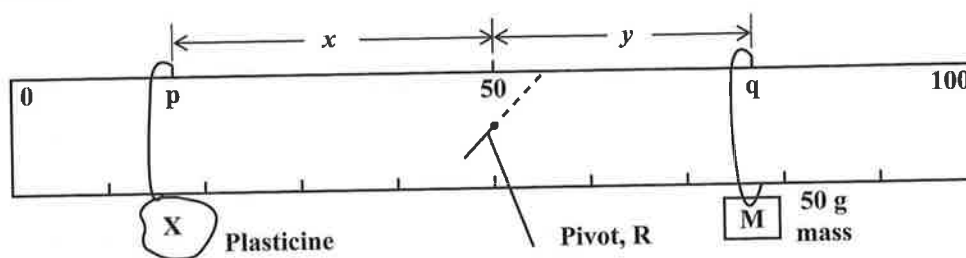


Figure 1. Setup of the activity

The results of the activity were recorded in Table 1 below.

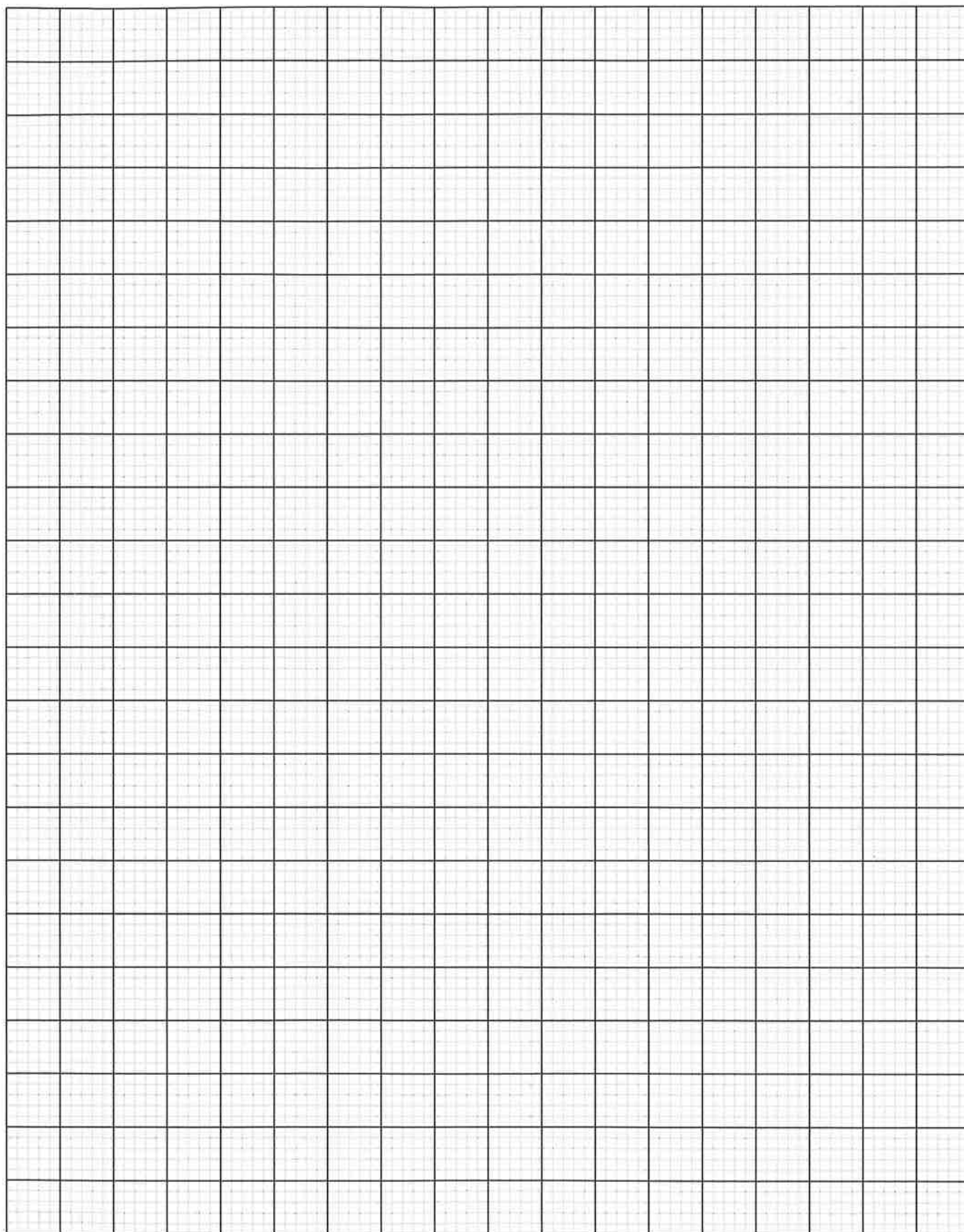
TABLE 1: RESULTS OF THE ACTIVITY

Position of Plasticine, p/cm	Position of 50 g mass q/cm	x/cm	y/cm
10.0	86.0	40.0	
20.0	76.2	30.0	
30.0	66.0		16.0
35.0	62.8		12.8
40.0	57.9	10.0	
45.0	53.6	5.0	

- (a) Complete Table 1 by calculating the values of x and y .

(3 marks)

GO ON TO THE NEXT PAGE



- (b) Use the results from the completed Table 1 to plot a graph of y/cm against x/cm , on the graph paper provided on page 3. **(6 marks)**

- (c) From your graph, calculate the slope, z .

(5 marks)

- (d) Use your graph to determine how far the 50 g mass needs to be placed if it balances when the plasticine was placed 27.5 cm from the pivot, R .

(1 mark)

- (e) State the Principle of Moments.

(4 marks)

- (f) The weight of the plasticine, W_p , is related to the weight of the 50 g mass, W_m , by the formula,

$$W_p = \text{gradient} \times W_m.$$

- (i) Determine the weight of the plasticine, W_p .

(4 marks)

GO ON TO THE NEXT PAGE

- (ii) Calculate the mass of the plasticine, M_p .

$[g = 10 \text{ N kg}^{-1}]$

(2 marks)

Total 25 marks

2. (a) (i) Complete Table 2 relating fundamental quantities and their base unit symbols.

TABLE 2: FUNDAMENTAL QUANTITIES AND THEIR BASE UNITS

Name	Symbol	Base (SI) Unit
Mass	m	kg
Time		
Current		
Temperature		
Length		

(4 marks)

- (ii) State the difference between a 'linear scale' and a 'non-linear scale'.

(2 marks)

- (iii) You are provided with the following three measuring instruments.

Metre rule, Clinical thermometer, Conical flask

Which of the instruments listed above has a non-linear scale?

(1 mark)

- (b) The container shown in Figure 2 is completely filled with cement.

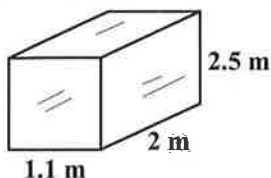


Figure 2. Container filled with cement

- (i) If the mass of the cement is 15 000 kg, calculate the density of this cement sample in kg m^{-3} .

(4 marks)

- (ii) What pressure, in Newtons, would the cement exert on the base of the container?

[$g = 10 \text{ N kg}^{-1}$]

(4 marks)

Total 15 marks

3. (a) (i) Albert Einstein put forward a theory that if the energy of a body changes, then its mass also changes. State the equation which relates these changes, clearly identifying the symbols or letters used in the equation.

(3 marks)

- (ii) State TWO arguments EACH **for** and **against** the utilization of nuclear energy.

For:

1.

2.

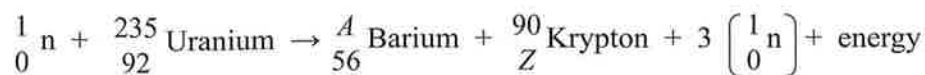
Against:

1.

2.

(4 marks)

- (b) Nuclear fission is a nuclear reaction in which a moving neutron strikes the nucleus of an atom splitting it into lighter nuclei, often producing free neutrons and photons in the form of gamma rays. One such reaction is given below.



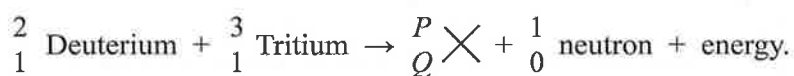
- (i) Calculate the values of A and Z .

(2 marks)

- (ii) The energy released in the given reaction is 1.8×10^{-12} J. Calculate the excess mass (mass change) in this reaction.

(3 marks)

- (c) Fusion is the joining together of two nuclei. One possible fusion reaction is



By calculating P and Q , identify the nucleus X.

$$[c = 3.0 \times 10^8 \text{ ms}^{-1}]$$

(3 marks)

Total 15 marks

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SECTION B

Attempt ALL questions.

You MUST write your answers in the spaces provided after each question.

4. (a) Figure 3 shows the diagram of a simple d.c. motor connected in a circuit.

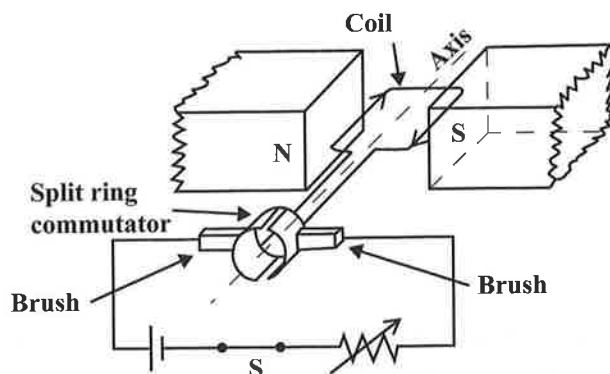


Figure 3. D.C. motor

With reference to the d.c. motor shown in Figure 3, explain how the coil is able to rotate continuously. (6 marks)

- (b) The waveform shown in Figure 4 depicts the current through a resistor connected in an a.c. circuit.

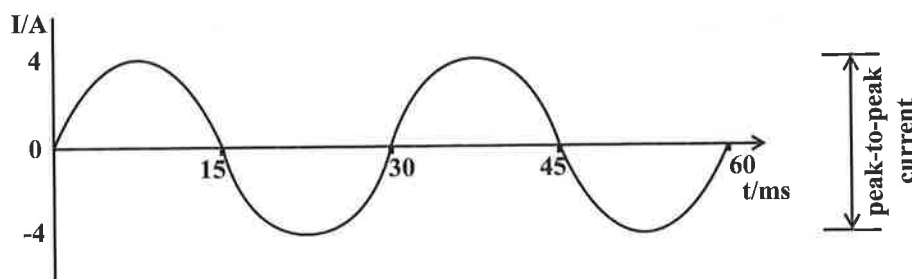


Figure 4. A.C. waveform

- (i) Calculate the period and frequency of the alternating supply. (3 marks)
- (ii) If the resistor is 45Ω , determine the peak-to-peak voltage across the resistor. (3 marks)
- (iii) Calculate the maximum power dissipated in the resistor. (3 marks)

Total 15 marks

GO ON TO THE NEXT PAGE

5. (a) Figure 5 shows two blocks of copper that are placed in contact with each other. Block *A* initially has a temperature of 40°C and block *B* has a temperature of 70°C .

BLOCK A 40°C
BLOCK B 70°C

Figure 5. Two blocks of copper

- (i) What is the direction of net heat transfer between the blocks? **(1 mark)**
- (ii) The final equilibrium temperature of the blocks is 52°C . Convert this temperature to Kelvin. **(3 marks)**
- (b) Figure 6 shows the sketch of a natural convection solar water heating system.

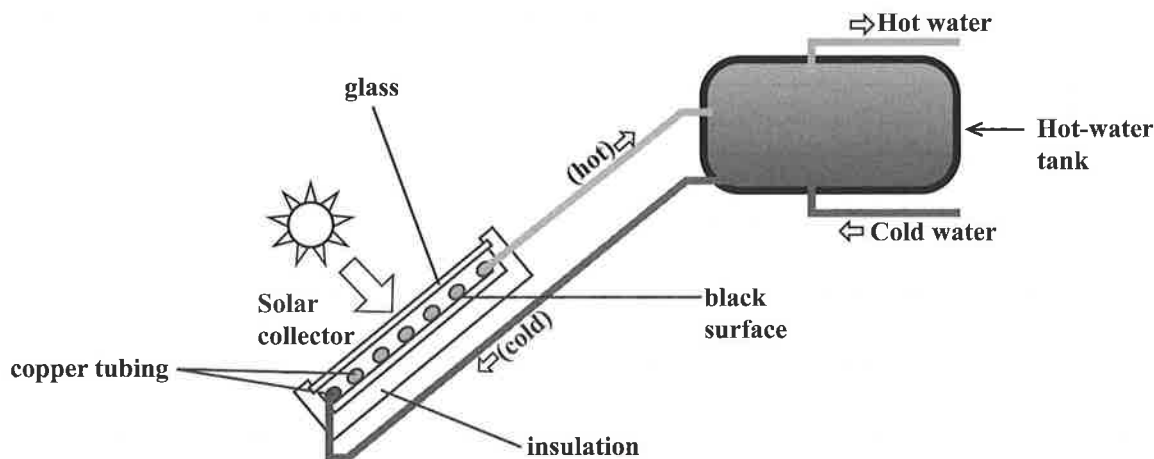


Figure 6. Solar water heating system

- (i) The energy absorbed by the solar collector is 300 kJ and 40% of this is transferred to the 2 kg of water in the copper tubing. If the initial temperature of the water is 27°C , calculate the final temperature of the water. **(6 marks)**

- (ii) Outline which feature of the solar water heating system shown in Figure 6 demonstrates EACH of the following thermal energy principles:
- a) Good absorption of heat
 - b) Heat transfer by convection
 - c) The green house effect
 - d) Good heat transfer by conduction
 - e) Reduction of heat transfer by conduction

[Specific heat capacity of water = $4.2 \times 10^3 \text{ J kg}^{-1} \text{ K}^{-1}$]

(5 marks)

Total 15 marks

Write your answer to Question 5 here.

This image shows a single sheet of white paper with horizontal blue or grey ruling lines. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

6. (a) (i) What is the difference between 'longitudinal waves' and 'transverse waves'?
(2 marks)
- (ii) Given that red light has a wavelength of 700×10^{-9} m, calculate its frequency.
(4 marks)
- (b) In 2010, Sound Navigation and Ranging (SONAR) was used to detect the presence of submerged oil plumes in the Gulf of Mexico after a massive oil spill. Figure 7 shows a ship using SONAR to locate an oil plume.

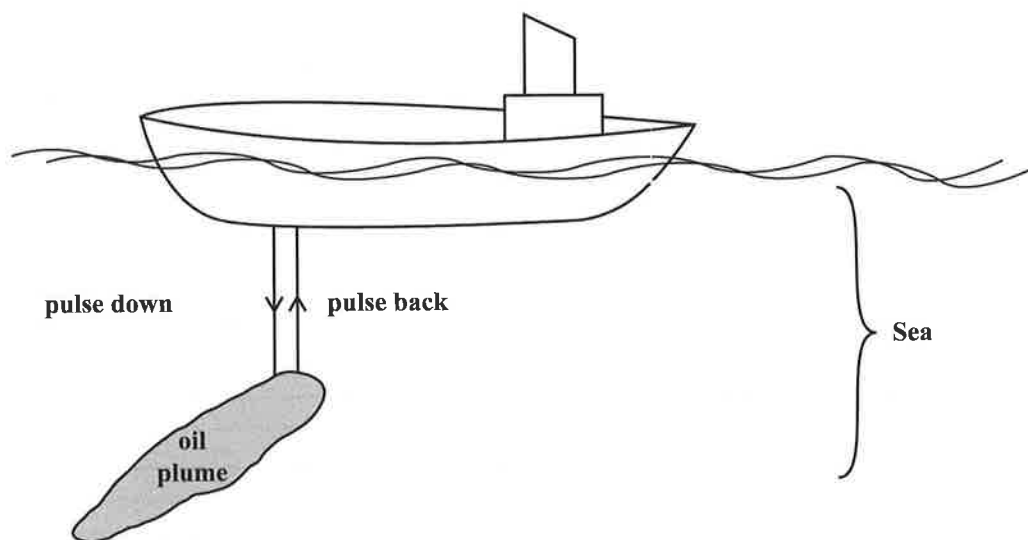


Figure 7. SONAR ship

- (i) Explain, with reference to Figure 7, how SONAR was used to determine the depth of an oil plume.
(4 marks)
- (ii) If echoes are received 0.3 seconds after being sent, calculate the depth of the oil plume below the detector, given that $\text{Velocity} = \frac{\text{distance}}{\text{time}}$.

[Speed of sound in sea water = 1450 ms^{-1}]

[Speed of light in air = $3.0 \times 10^8 \text{ ms}^{-1}$]

(5 marks)

Total 15 marks

